

X-37B Space Plane's Microwave Power Beam Experiment Is A Way Bigger Deal Than It Seems

When the X-37B launches on May 16, it will carry a technology that could eventually allow drones to stay aloft indefinitely anywhere on the globe.

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UPDATED MAY 13, 2020 8:04 PM EDT



The shadowy X-37B, the Air Force's unmanned, reusable spacecraft, is set to launch for its sixth flight on May 16 from Cape Canaveral Air Force Station, Florida. While most of the payloads set for the flight are standard fare for space experiments, at least the ones that are disclosed, one of them has immense potential implications for the future of remote power generation and especially long-endurance unmanned aircraft propulsion.

The X-37B's upcoming mission is known as both Orbital Test Vehicle-6 (OTV-6) and U.S. Space Force-7 (USSF-7). It will carry out missions that will assess the effects of cosmic radiation and other "space effects" on plant seeds and various material samples. According to a Space Force press release, which went out on May 6, another payload aboard the X-37B will be an experimental system designed by the Naval Research Laboratory that is capable of capturing solar power and beaming that energy back to Earth in the form of microwaves.

While the press releases of the Department of Defense and the Space Force are scant on details, the Naval Research Laboratory's head of beamed power has explicitly stated in the past that this system has enormous implications when it comes to long-endurance unmanned aerial vehicles (UAVs). In addition, it could allow satellites to provide reliable power anywhere on the planet or even to spacecraft or other satellites in orbit.



The U.S. Naval Research Laboratory (NRL) has been promoting similar-sounding technologies over the last year. In October 2019, the NRL conducted a three-day long demonstration of the Navy's latest power-beaming capabilities at the Naval Surface Warfare Center in Bethesda, Maryland. In the demonstration, the NRL transmitted a completely silent, invisible beam of 2-kilowatt laser power over 300 meters through the air over attendees' heads. The demonstration was meant to showcase the safety and technological readiness of this utterly transformative concept.

As far as the aforementioned applications of this capability, research or military outposts in remote locations would no longer have to rely on low-power solar systems or haul heavy generators and large amounts of fuel, but instead could bring a rectifying antenna, or rectenna, to capture energy in the form of microwaves beamed from satellites overhead. In addition, areas ravaged by natural disasters could use the system to generate electricity to aid in reconstruction efforts long before traditional electrical infrastructure is rebuilt. It could even power autonomous ships at sea.

[...]

Beamed Power And The Future of UAVs

In 2014, the superintendent of the Naval Research Laboratory's Plasma Physics Division Thomas Mehlhorn published a paper in IEEE Transactions on Plasma Sciences which offered an overview of plasma physics and pulsed power as they relate to national security. The article spans a wide variety of topics including nuclear weapons, inertial confinement fusion, and high-energy laser weapons. In the paper, Mehlhorn also touches upon the Navy's beamed power UAV research at the time, writing that the continuous flight times offered by beamed power systems could change surveillance, reconnaissance, and communications gateway/relay missions forever:

"Building upon the concept of scalability, rather than using a laser beam to kill a UAV, they began to pursue the idea of beaming power to a UAV to allow continuous flight, with potential application to both surveillance [Intelligence, Surveillance, and Reconnaissance (ISR)] and countermeasure missions. The team has pursued this idea using NRL applied research funds with the vision that long-range laser power beaming to UAVs could allow for long-duration flights with reduced manpower requirements for many Navy and DoD missions, including off-board decoys, persistent surveillance, and communication relays."

According to an October 2019 Navy.mil press release, the Navy's beamed power system has also been endorsed by the Marines, Army, and Air Force and is expected to throughout the Department of Defense in the near future. The extent to which such systems have already been tested or deployed is unclear, although the Department of Energy has explored the concept of beaming microwaves from space since at least 2014. Doing the same from the ground, within line of sight of the aircraft, which can

still be dozens or even hundreds of miles away depending on the altitudes involved, is such an easier task that it would be a bit puzzling if the technology isn't already under development, or even possibly in some sort of clandestine operational state.

The NRL program is being funded through the Operational Energy Capability Improvement Fund (OECIF), which "incentivizes S&T to promote long term change in U.S. Department of Defense capabilities" and "fosters innovation to improve operational energy performance."

Doing so from another aircraft is also clearly an objective based on the existing literature and would help mitigate the line of sight limitations with ground-based power beaming stations, but would sacrifice endurance and simplicity. In the 2011 RAND study cited above, the authors write that possibilities for beamed power applications include "ultra-high-altitude observation stations or communication relays and flocks of high-altitude sensor probes powered remotely from a large aircraft 'mother ship.'"

Meanwhile, the China Academy of Space Technology claimed to already be testing such a system in 2019 and said that a fully-functional Chinese microwave beaming power station in space could be deployed by 2050.

As you can probably tell at this point, this technology has massive implications not only for the future of UAVs, but for all of mankind. Such a system could be used to keep UAVs in the air for very long periods of time to replace cell towers or communications satellites in the event of a crisis in a region or even for normal operations of increasingly complex communications networks. Unlike a tethered aerostat, these UAVs would require far less infrastructure, could be moved around at will for optimum coverage, and could land quickly for servicing. They could even deploy dozens of miles, or even further, away from their base stations. With a space-based power source, they could fly anywhere on earth. Obviously, the implications for overhead surveillance are equally impactful.

So, while the X-37B's latest mission details seem neat on a scientific level, the reality is the microwave system it is testing could change the game for many military-related applications and could actually open the door for near-continuous unmanned flight throughout the atmosphere.

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Source: <https://www.twz.com/33339/x-37b-space-planes-microwave-power-beam-experiment-is-a-way-bigger-deal-than-it-seems>
